APC Identification Problem: Simuatlion Experiment based on real data of 16 narwhals

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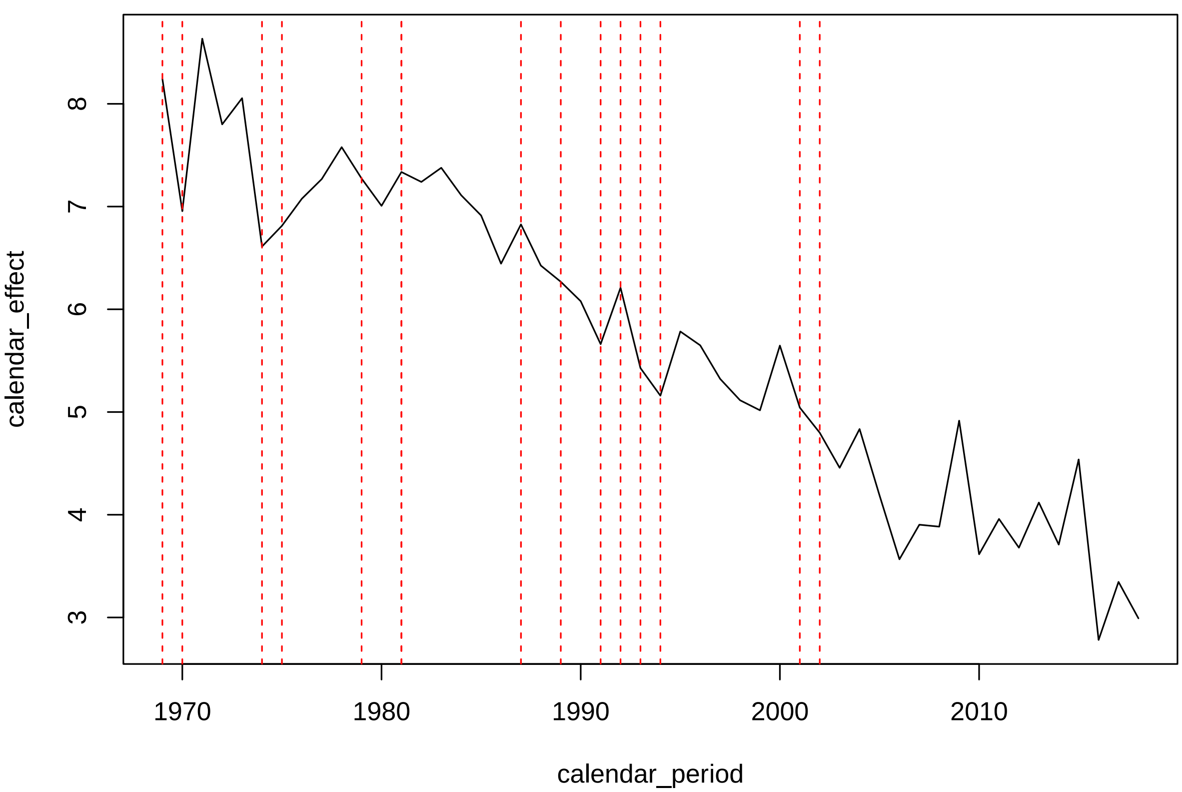
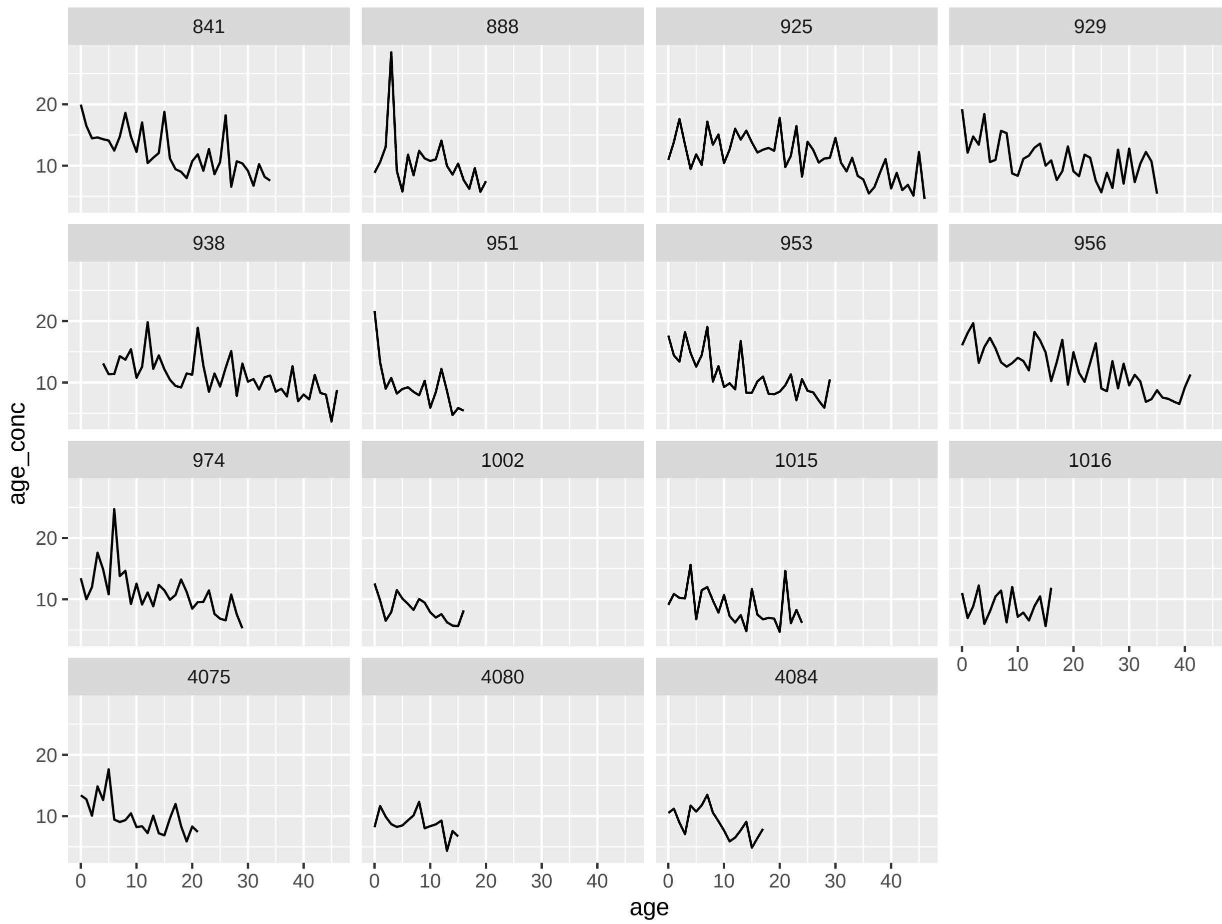
SIMULATION EXPERIMENT 1: Linear DECREASE in environmental concentration

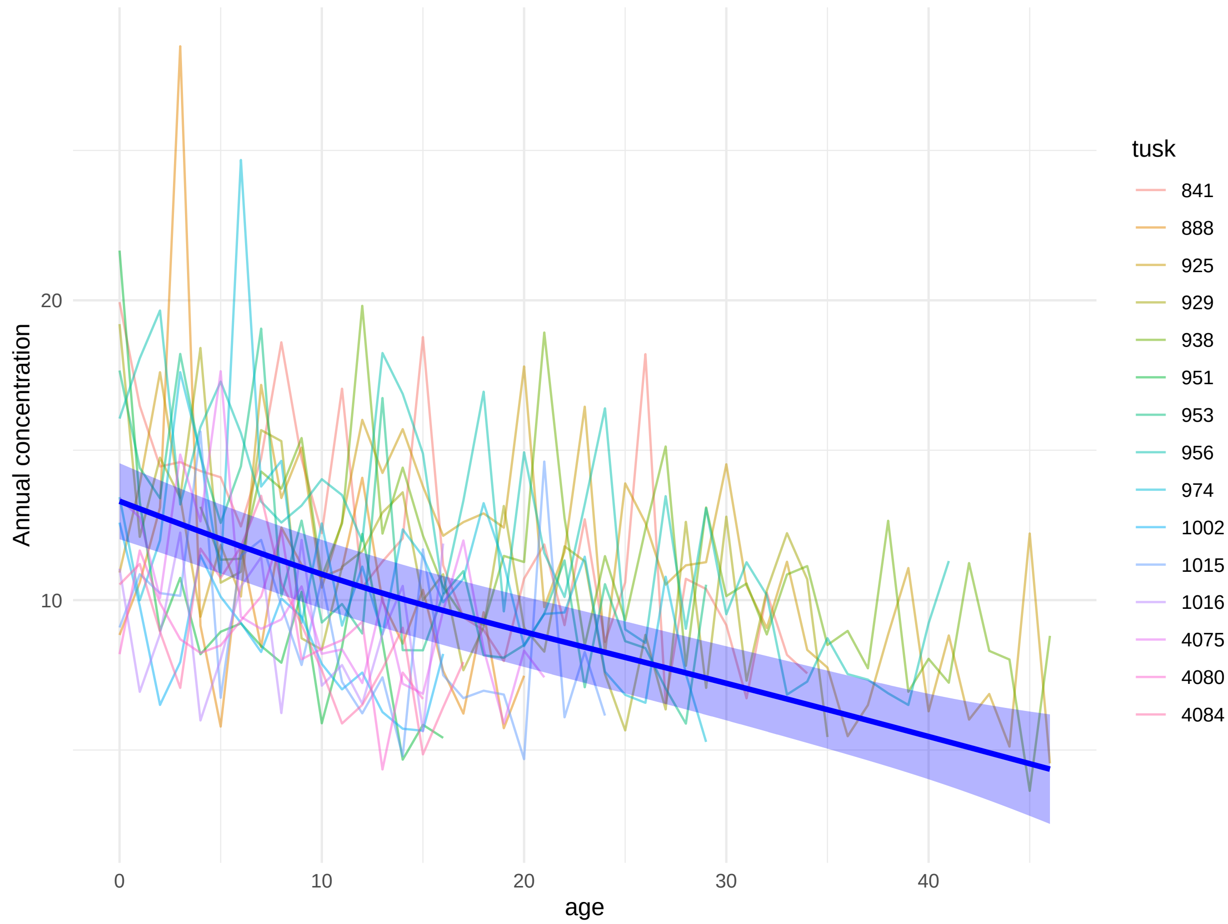
We start by simulating a climate (or environmental) signal describing the concentration of some element of interest in the arctic water. We assume a linear trend and gaussian noise. The dashed red lines are estimated year of birth of the (real) population of narwhals in the original dataset. Notice that age and year are estimated using a age estimation technique based on Growth Layer Groups and Laser Ablation derived signals from their tusks.



Assuming that each individual narwhal incorporates the element (in the water) at a constant rate - which is allowed to fluctuate (modelled as gamma noise) due to stress or illness - we can plot the observed concentration as a function of age. This is done below.

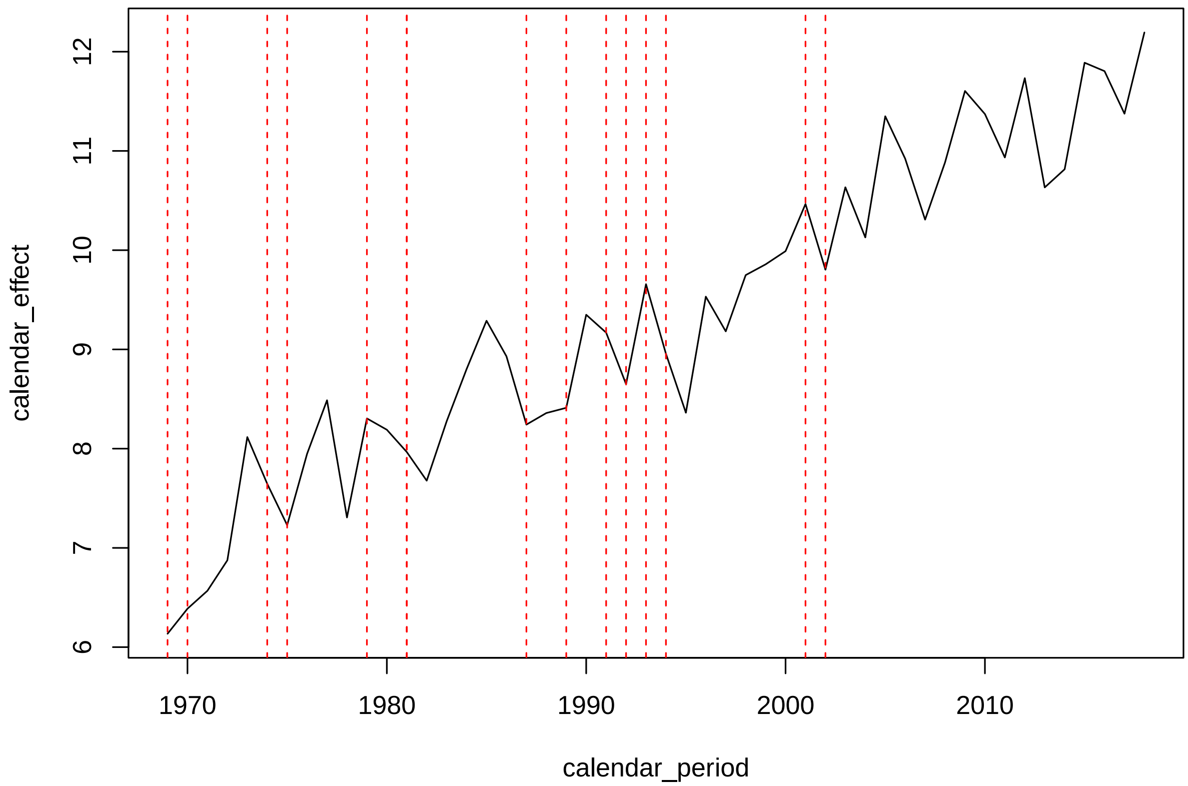


If we now plot all the concentrations in a common plot (all tusks) as a function of age, we see that the observations follows the trend of the environmental signal.



SIMULATION EXPERIMENT 2: Linear INCREASE in environmental concentration

We can now repeat the procedure above, but using a linear increasing climate signal. Still assuming constant deposition/uptake rate of elements in the narwhal tusk.



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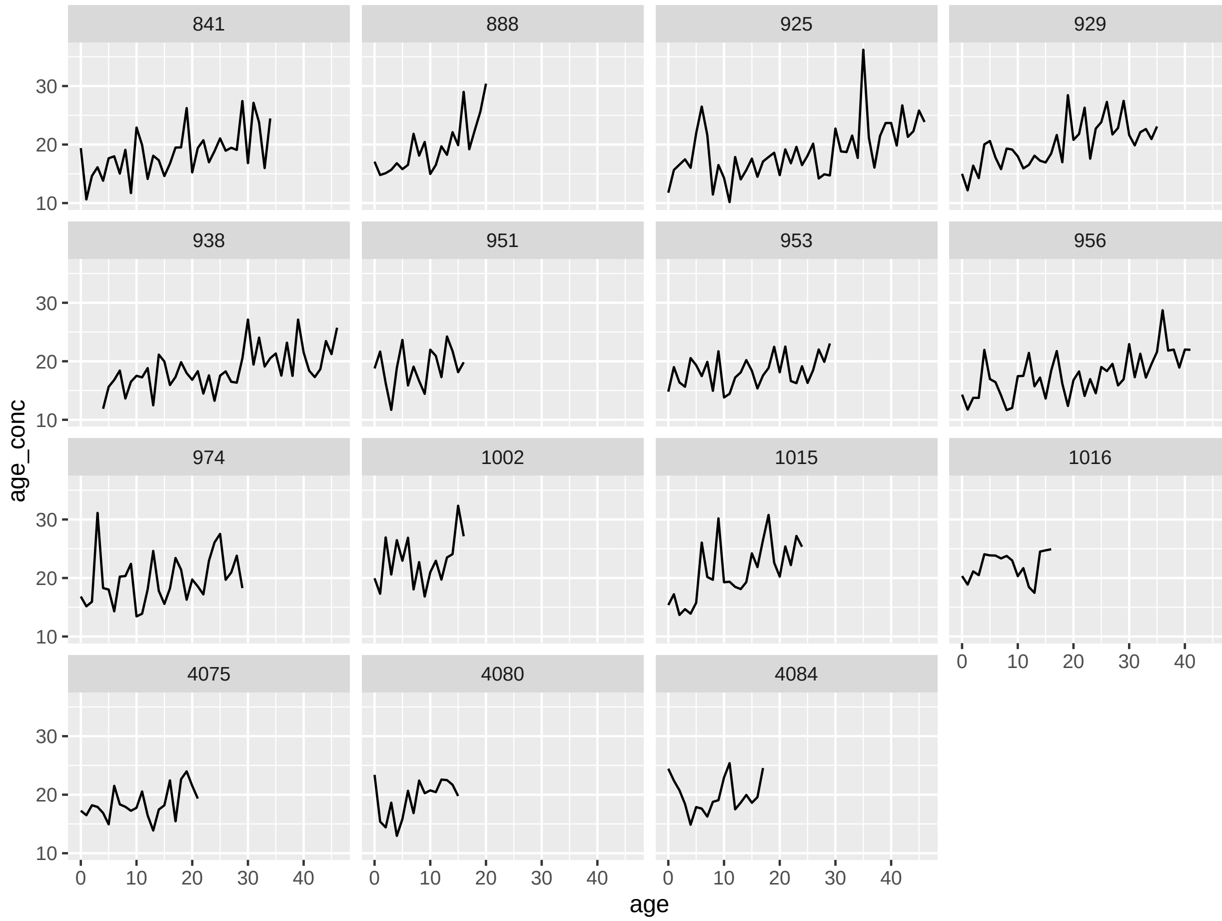
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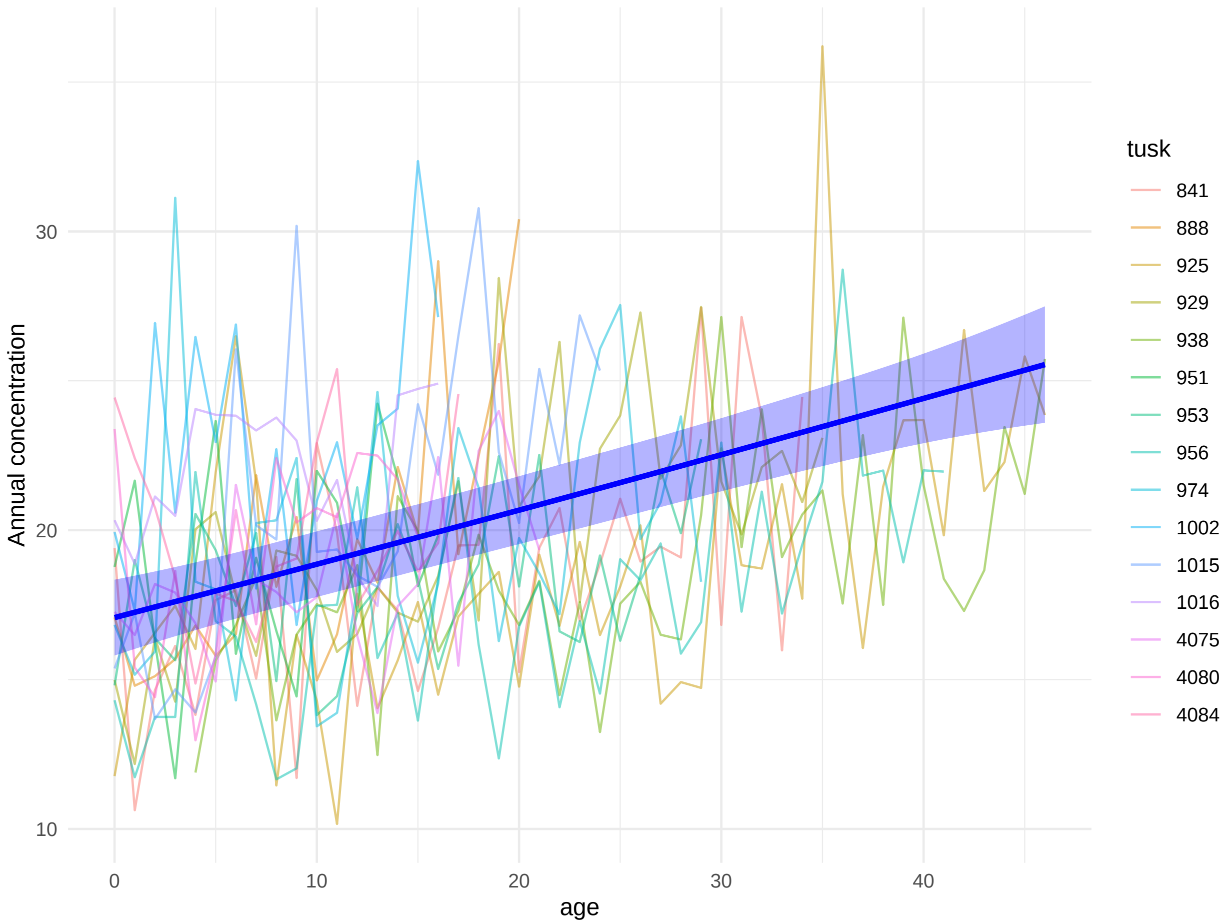
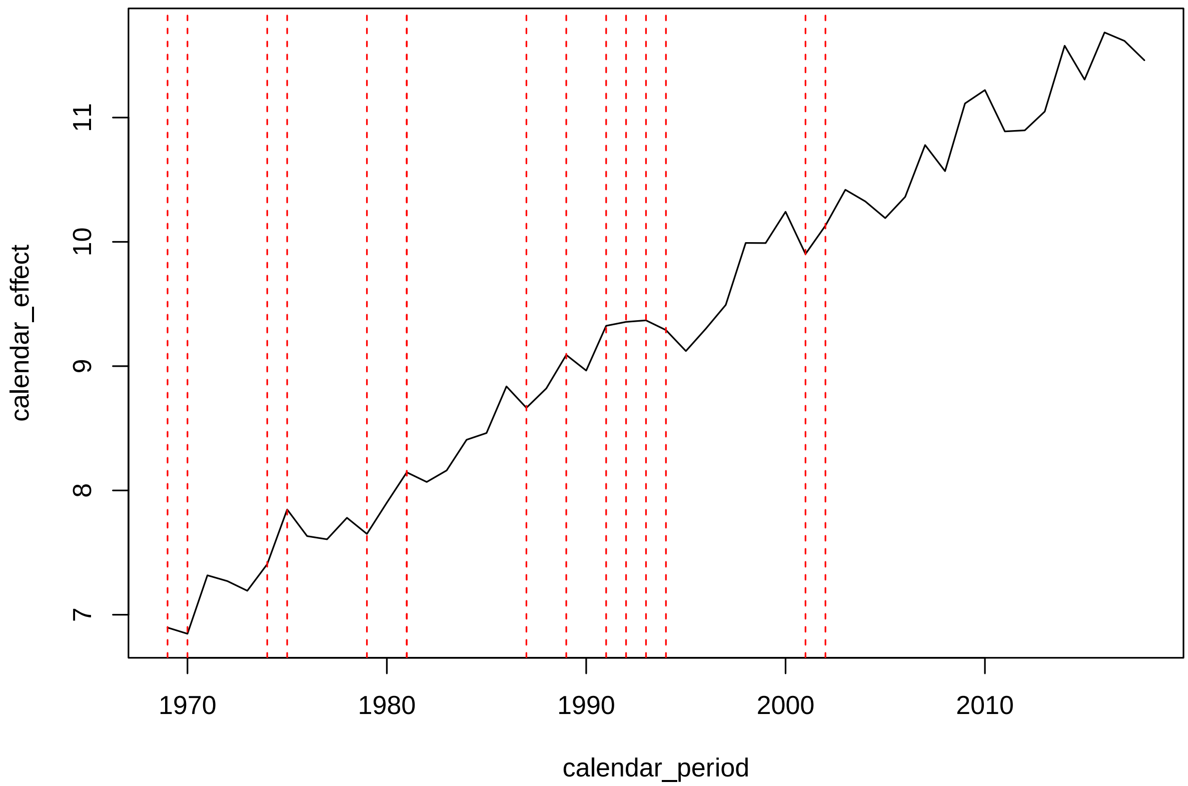


SIMULATION EXPERIMENT 3: Linear INCREASE in environmental concentration and exponential decay in individual incorporation

We again assume a linear increase in the environmental signal, but now assume a exponential decay in the deposition rate of elements. Notice how the combined (concentration,age)-plot now reflects the incorporation mechanism, i.e. the age component, rather than the climate signal, i.e. the year component.



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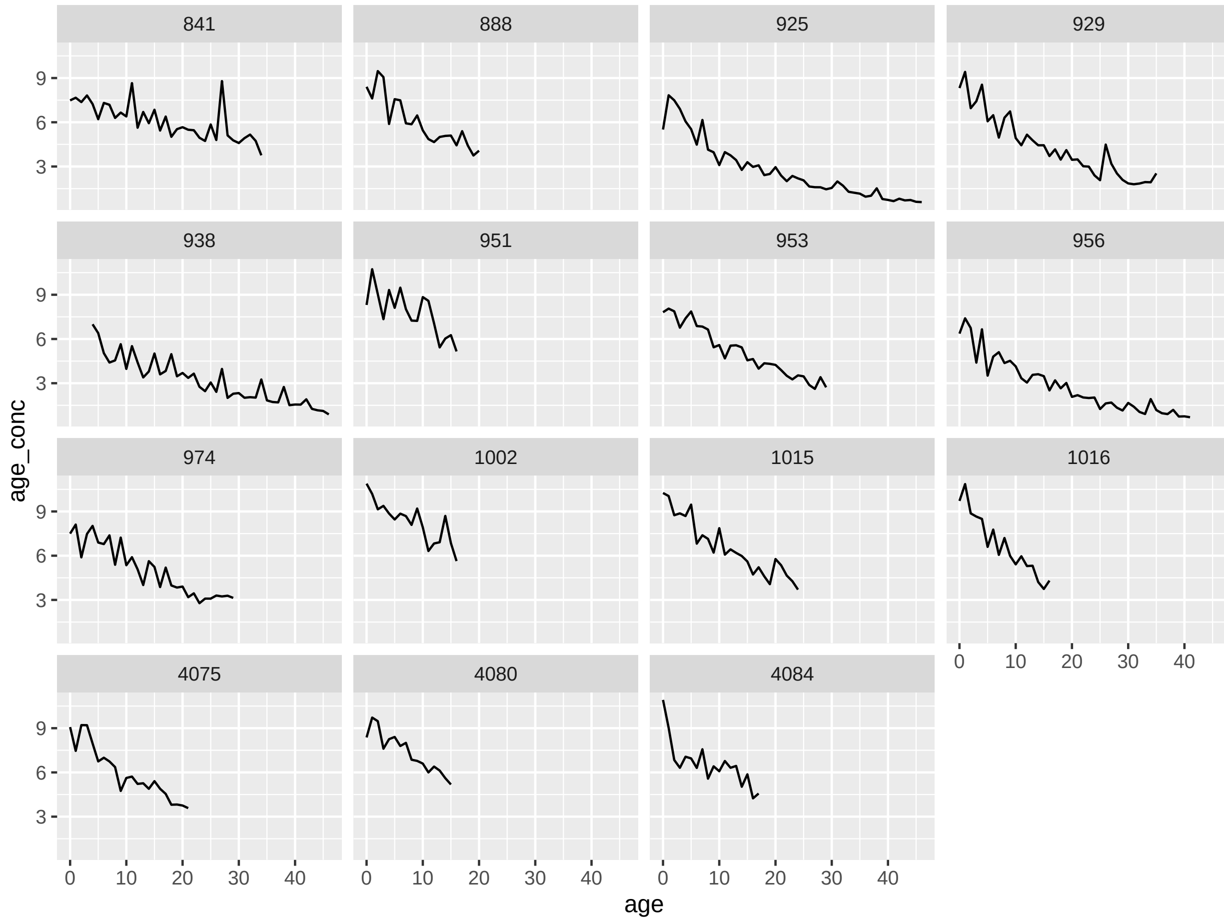
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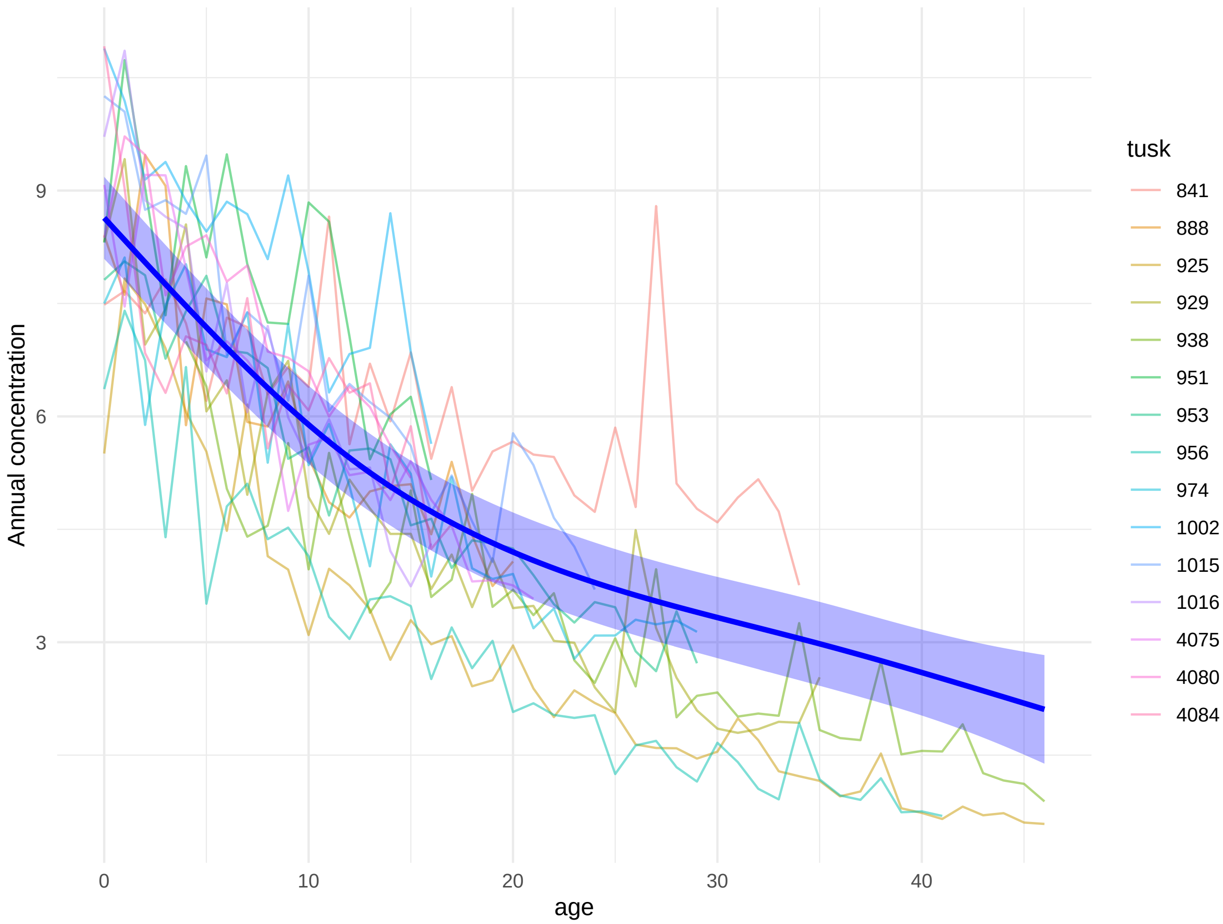
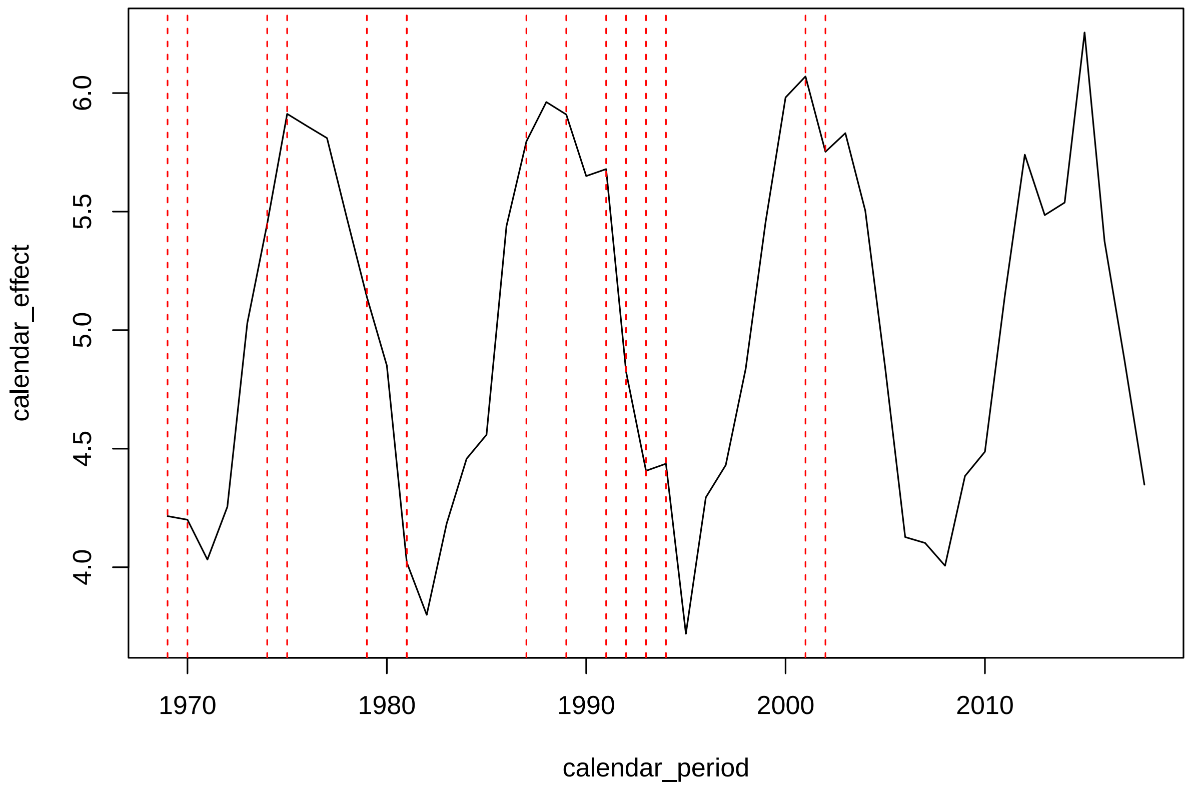


SIMULATION EXPERIMENT 4: Periodic change in environment concentration

Lets look at a periodic environmental signal, but assume constant deposition rate. In this case, the combined plot identifies the constant age component, since the observed signals of each whale with a periodic component tends to cancel out.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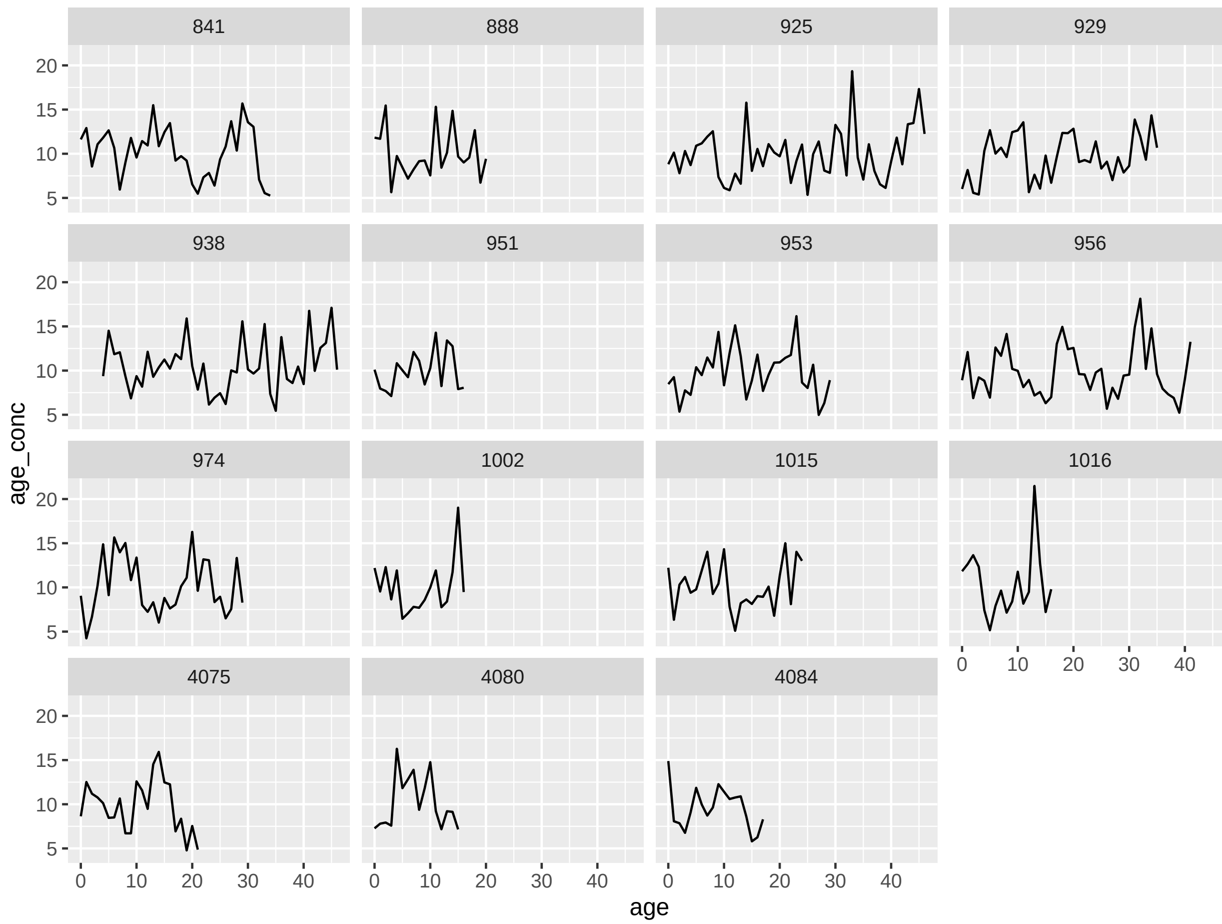
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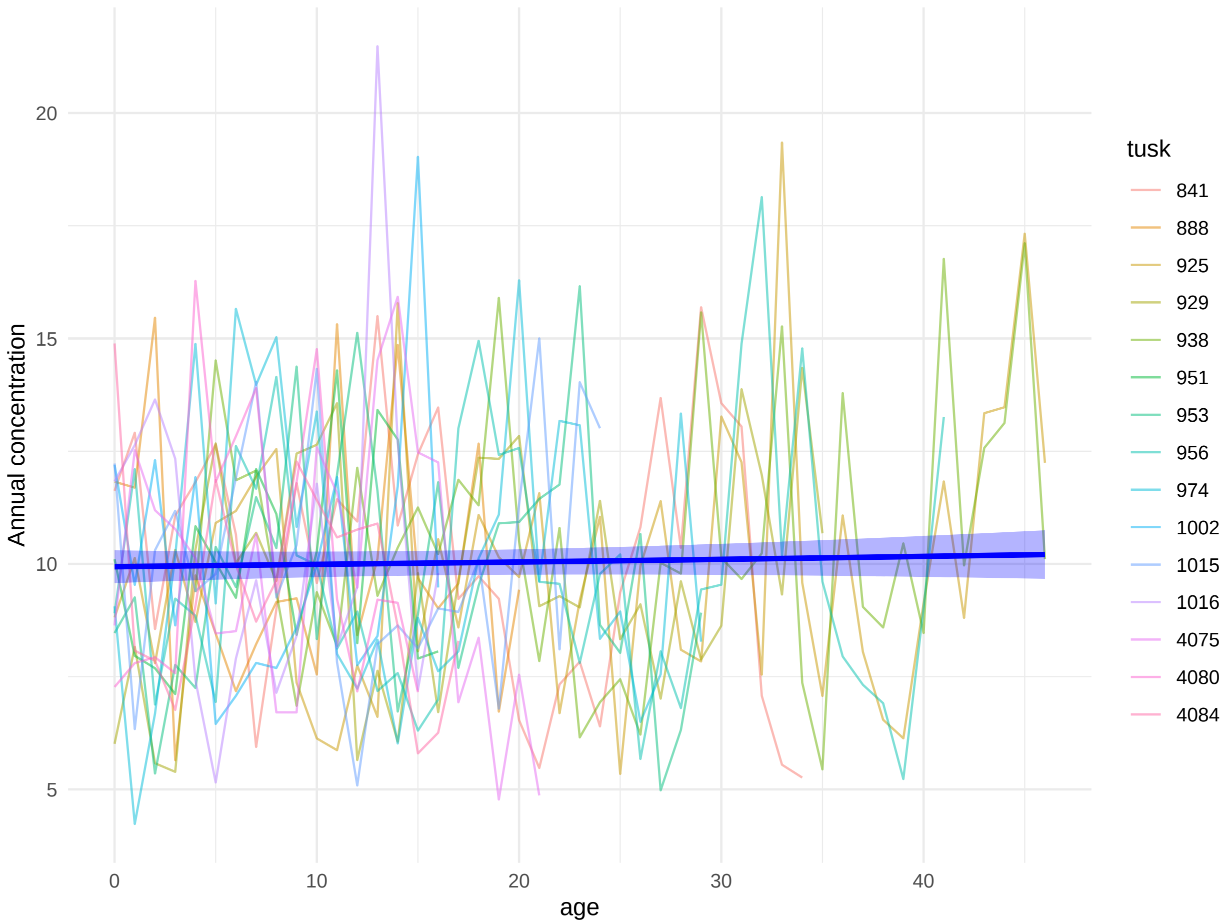
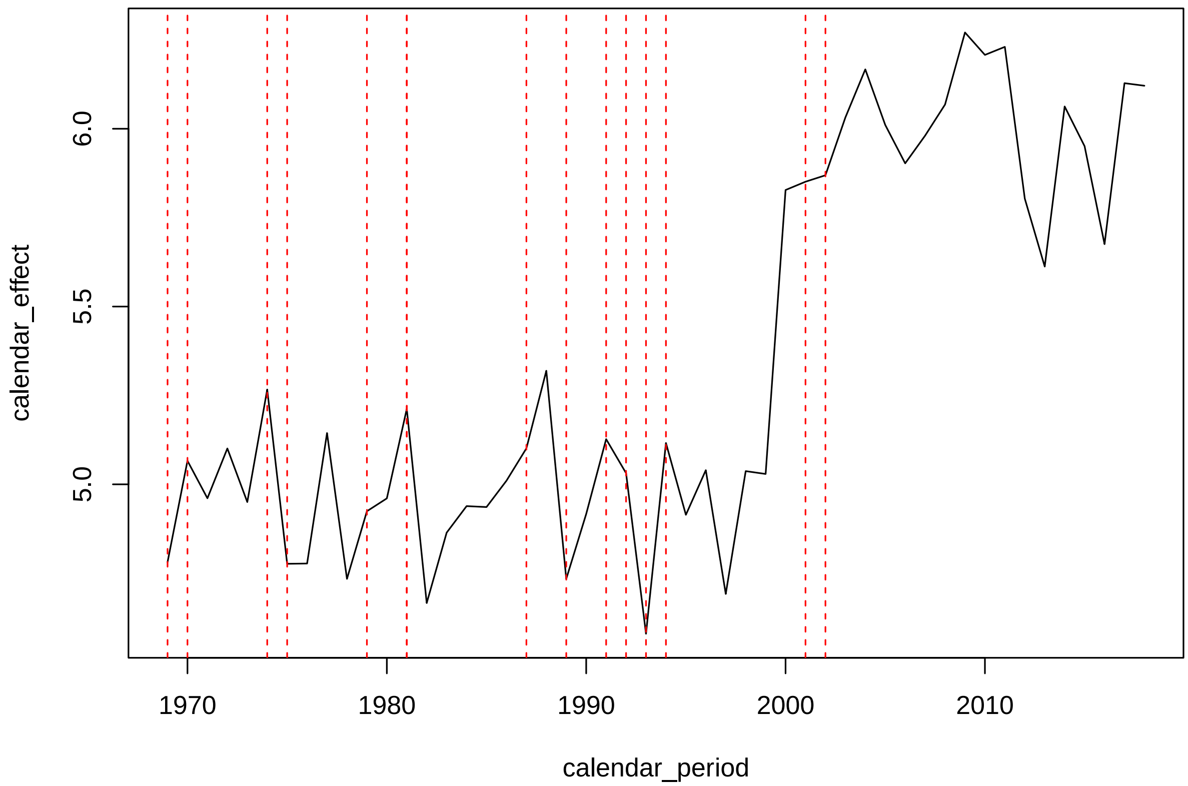


SIMULATION EXPERIMENT 5: Sudden change in environment concentration

Finally we consider a regime shift in the climate at around year 2000, but assume constant deposition rate. In this case, we observe a linearly increasing trend in the combined plot owing the level change in the climate signal.



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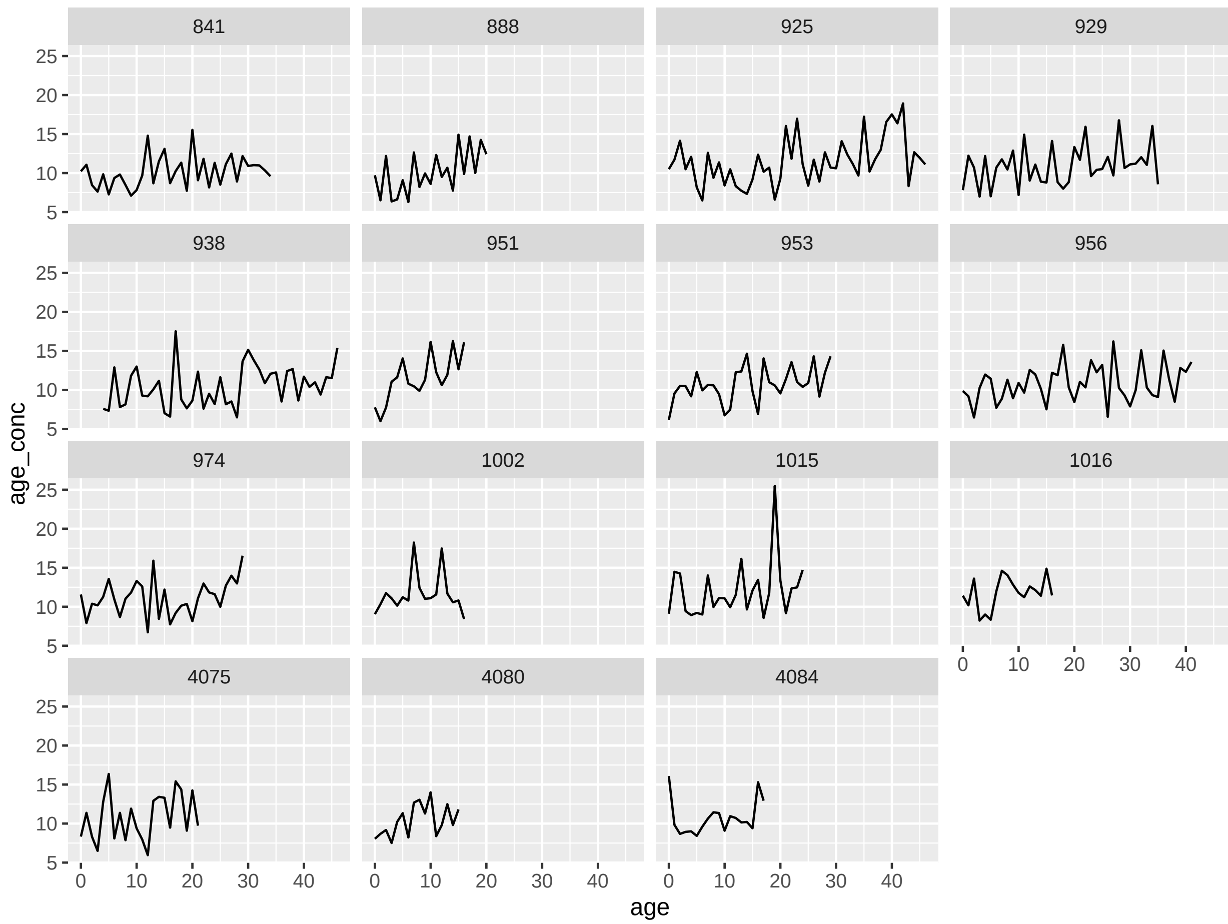
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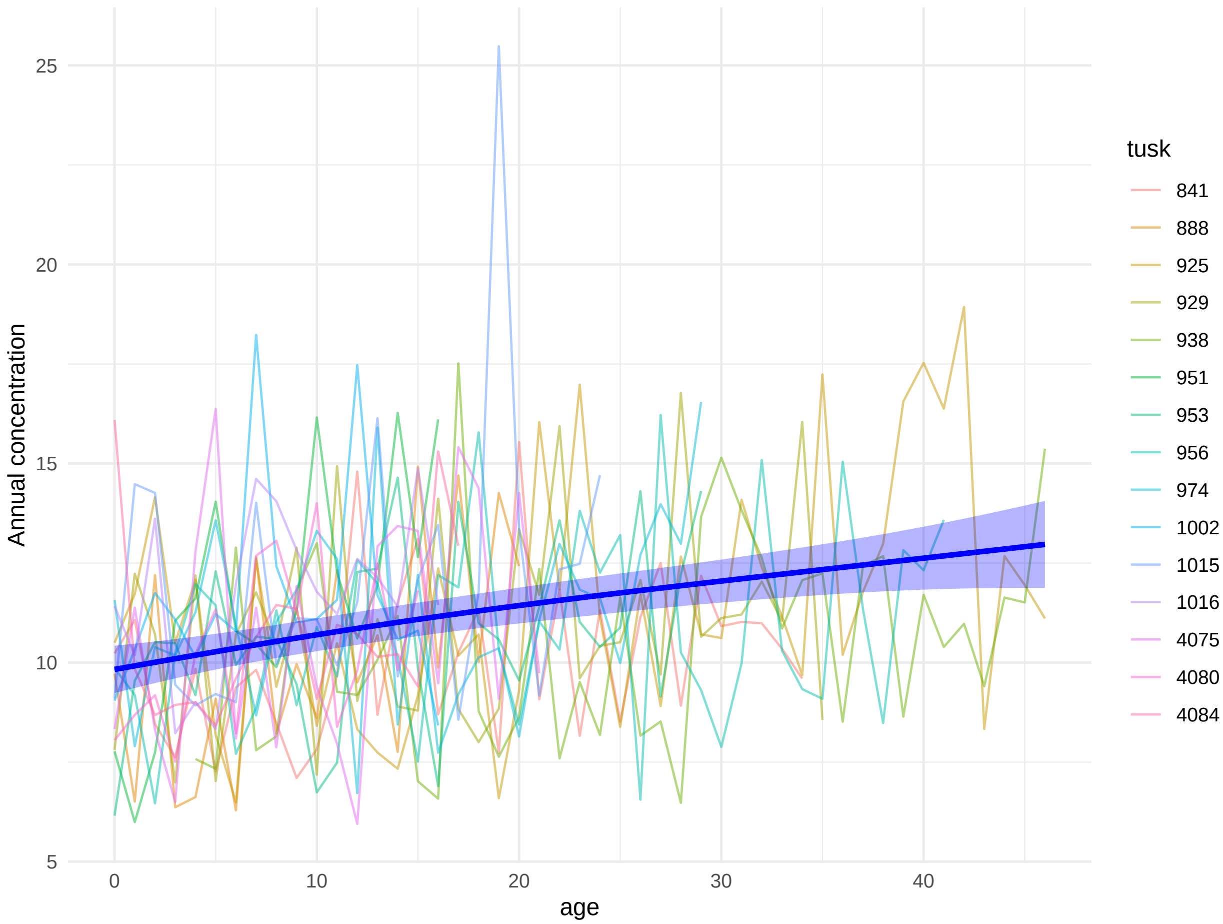

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